

# **Temporal vs. Spatial Visual Masking: Dissociable Cognitive Mechanisms and Links to Intelligence**

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# Intelligence and sensory discrimination

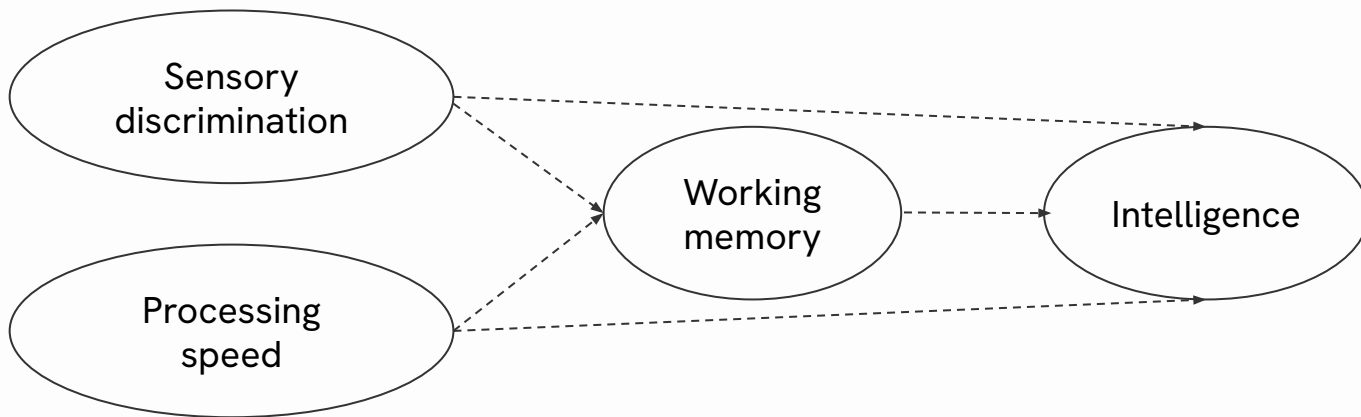
- Galton (1883): Individual differences in general sensory discrimination ability, such as sensory acuity in vision, hearing, and other senses, are strongly inter-correlated. Intelligence originates from the general sensory discrimination ability.
  - Conclusion supported by impressions and anecdotes, no statistics
- Wissler (1901): concluded that no reliable sensory difference can be used for detecting intelligence differences
- Spearman (1904), Thorndike (1909), Abelson (1911): with the advancement in psychometric and statistical analysis, they concluded a mild correlation of  $r = 0.2-0.4$  between sensory discrimination and intelligence (g factor).

# Intelligence and processing speed

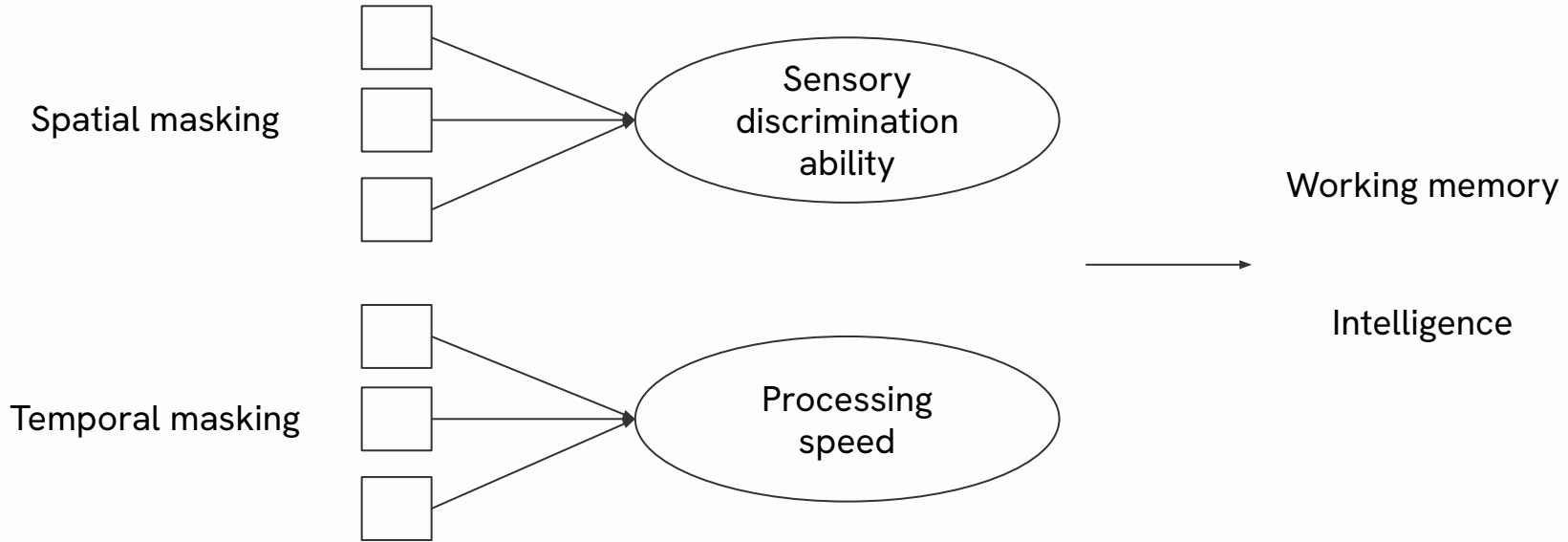
- Deary (1995): Auditory Inspection Time correlates with Intelligence ( $r \sim -0.4$ )
- Nettelbeck (1976): Visual Inspection Time correlates with Intelligence ( $r \sim -0.5$ )
- Grudnik and Kranzler (2001): Meta-analysis of the correlation between inspection time and intelligence ( $r \sim 0.5$  after correction,  $r \sim 0.3$  before correction)

# Working memory as a bottleneck?

- Jastrzębski (2020): WM may likely constitute a central bottleneck common to many sensorimotor tasks, because the stimuli need to be somehow represented mentally in order to be compared, and the participants' performance may depend on the quality of such a representation.
- Ackerman (2005): Meta-Analysis: Correlation between WM and g is  $r = 0.47$ .



# Use temporal vs. spatial visual masking to access processing speed and sensory discrimination ability

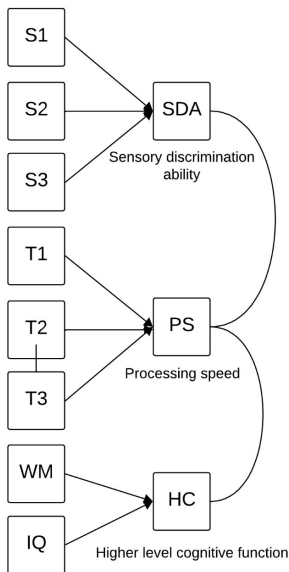


# Data analysis overview

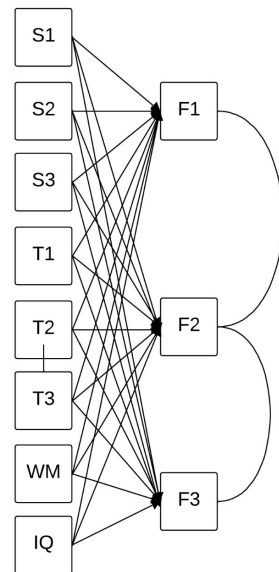
## Correlation

$$\begin{pmatrix} S_1 \\ S_2 \\ S_3 \\ T_1 \\ T_2 \\ T_3 \\ WM \\ IQ \end{pmatrix} \sim \mathcal{N}(\mu, \Sigma)$$

## Confirmatory



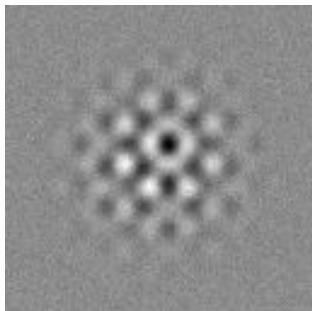
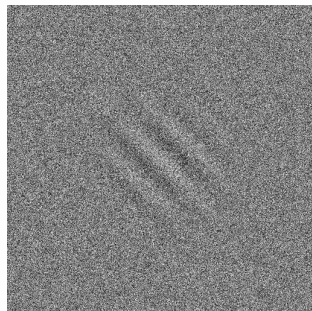
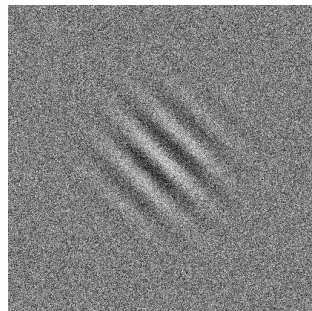
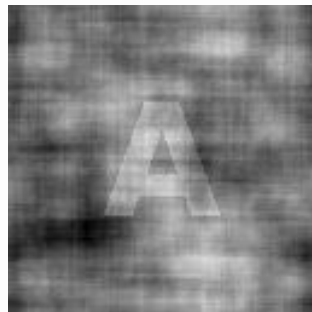
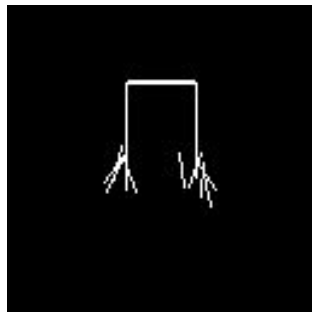
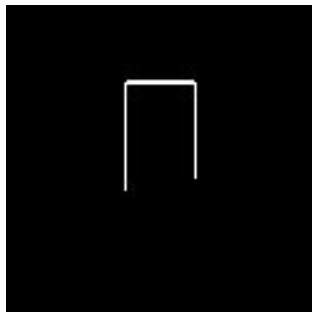
## Exploratory



# Tasks

|                  |        |                                                                                   |                                                                                                       |                                                                                 |
|------------------|--------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
|                  |        | Stimulus                                                                          | Spatial mask                                                                                          | Staircase adjustment                                                            |
|                  | Task 1 |  |                      | $\pm \Delta \text{Leg length}$<br>$\pm \text{Transparency of the random lines}$ |
| Spatial masking  | Task 2 |  |                      | $\pm \text{Noise}\%$                                                            |
|                  | Task 3 |  |                      | $\pm \text{Noise}\%$                                                            |
|                  |        | Stimulus                                                                          | Temporal mask                                                                                         | Staircase adjustment                                                            |
| Temporal masking | Task 4 |  | <br>$\pm \text{SOA}$ | $\pm \text{SOA}$                                                                |
|                  | Task 5 |  |                     | $\pm \text{SOA}$                                                                |
|                  | Task 6 |  |                      | $\pm \text{SOA}$                                                                |
| Working memory   | ?      |                                                                                   |                                                                                                       |                                                                                 |
| Intelligence     | ?      |                                                                                   |                                                                                                       |                                                                                 |

## Bigger pictures



# Questions

We are doing all 6 tasks in Psychopy, 165HZ refresh rate

- What are some suitable working memory tasks for this study
  - OSPAN, CSPAN?
- What are some suitable intelligence tasks for this study
  - Raven's Progressive Matrices and Sandia Matrices?
- Is there any platform/software that we can use directly?